

WEEK 3 CYBERSECURITY PROJECT

Password Cracking with John the Ripper (JTR)

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Module: Week 3 — Project Module 1

1. Project Overview

An authorized cybersecurity training exercise using John the Ripper and Johnny to recover the password of a protected PDF supplied for the lab.

2. Methodology

Configured Johnny with john.exe, extracted the PDF hash using the lab workflow, saved it as hash1.txt, loaded the hash into Johnny, and started the controlled password-recovery attack.

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John the Ripper password cracker

John the Ripper is an Open Source password security auditing and password recovery tool available for many operating systems. **John the Ripper jumbo** supports hundreds of hash and cipher types, including for: user passwords of Unix flavors (Linux, *BSD, Solaris, AIX, QNX, etc.), macOS, Windows, "web apps" (e.g., WordPress), groupware (e.g., Notes/Domino), and database servers (SQL, LDAP, etc.); network traffic captures (Windows network authentication, WiFi WPA-PSK, etc.); encrypted private keys (SSH, GnuPG, cryptocurrency wallets, etc.); filesystems and disks (macOS .dmg files and "sparse bundles", Windows BitLocker, etc.); archives (ZIP, RAR, 7z), and document files (PDF, Microsoft Office's, etc.) These are just some of the examples - there are many more.

[Openwall wordlists collection for password cracking \(20+ languages\)](#)

John the Ripper is free and Open Source software, distributed primarily in source code form. If you would rather use a commercial product, please consider [John the Ripper Pro](#), which is distributed primarily in the form of "native" packages for the target operating systems and in general is meant to be easier to install and use while delivering optimal performance.

Proceed to [John the Ripper Pro](#) homepage for your OS:

- [John the Ripper Pro for Linux](#)
- [John the Ripper Pro for macOS](#)
- [On Windows, consider Hash Suite](#) (developed by a contributor to John the Ripper)
- [On Android, consider Hash Suite Droid](#)

Download the latest John the Ripper development snapshot or jumbo release ([release notes](#)):

- Development source code in [GitHub repository](#) (download as [tar.gz](#) or [zip](#)) ← **recommended in preference of the release**
- 1.9.0-jumbo-1 sources in [tar.xz](#), 33 MB (signature) or [tar.gz](#), 43 MB (signature) ← **out of date, with known issues**
- 1.9.0-jumbo-1 64-bit Windows binaries in [7z](#), 22 MB (signature) or [zip](#), 63 MB (signature)
- 1.9.0-jumbo-1 32-bit Windows binaries in [7z](#), 21 MB (signature) or [zip](#), 61 MB (signature)

Run John the Ripper jumbo in the cloud (AWS):

- [John the Ripper in the cloud homepage](#)

Download the latest John the Ripper core release ([release notes](#)):

- 1.9.0 core sources in [tar.xz](#), 8.6 MB (signature) or [tar.gz](#), 13 MB (signature)
- Development source code in [GitHub repository](#)

To verify authenticity and integrity of your John the Ripper downloads, please use our [GnuPG public key](#). You will most likely need to download a "Windows binaries" archive above. However, if you choose to download the source code instead (for a specific good reason), then please refer to these pages on [how to extract John the Ripper source code from the tar.gz and tar.xz archives](#) and [how to build \(compile\) John the Ripper core](#) (for jumbo, please refer to instructions inside the archive). You can also consider the [unofficial builds on the contributed resources list](#) further down this page.

Get John the Ripper apparel at 0-Day Clothing and support the project

Binary redistributables

The current version is 2.2.

Binaries 2.2 (CURRENT)

- Windows : [johnny_2.2_win.zip](#)
- OS X : [johnny_2.2_osx.zip](#)
- Linux : binaries are not provided, you should build the software from source (see <https://github.com/openwall/johnny/blob/v2.2/INSTALL> and Sources section).

Binaries 2.1 (OLD)

- Windows : [johnny_2.1_win.zip](#)
- OS X : [johnny_2.1_osx.zip](#)
- Linux : binaries are not provided, you should build the software from source (see <https://github.com/openwall/johnny/blob/v2.1/INSTALL> and Sources section).

Binaries 2.0 (OLD)

- Windows : [johnny_2.0_win.zip](#)
- OS X : [johnny_2.0_osx.zip](#)
- Linux : binaries are not provided, you should build the software from source (see <https://github.com/openwall/johnny/blob/v2.0/INSTALL> and Sources section).

Binaries 1.1 (OLD)

Johnny does not have shortcuts for system menu yet. So type 'johnny' in your terminal to start Johnny.

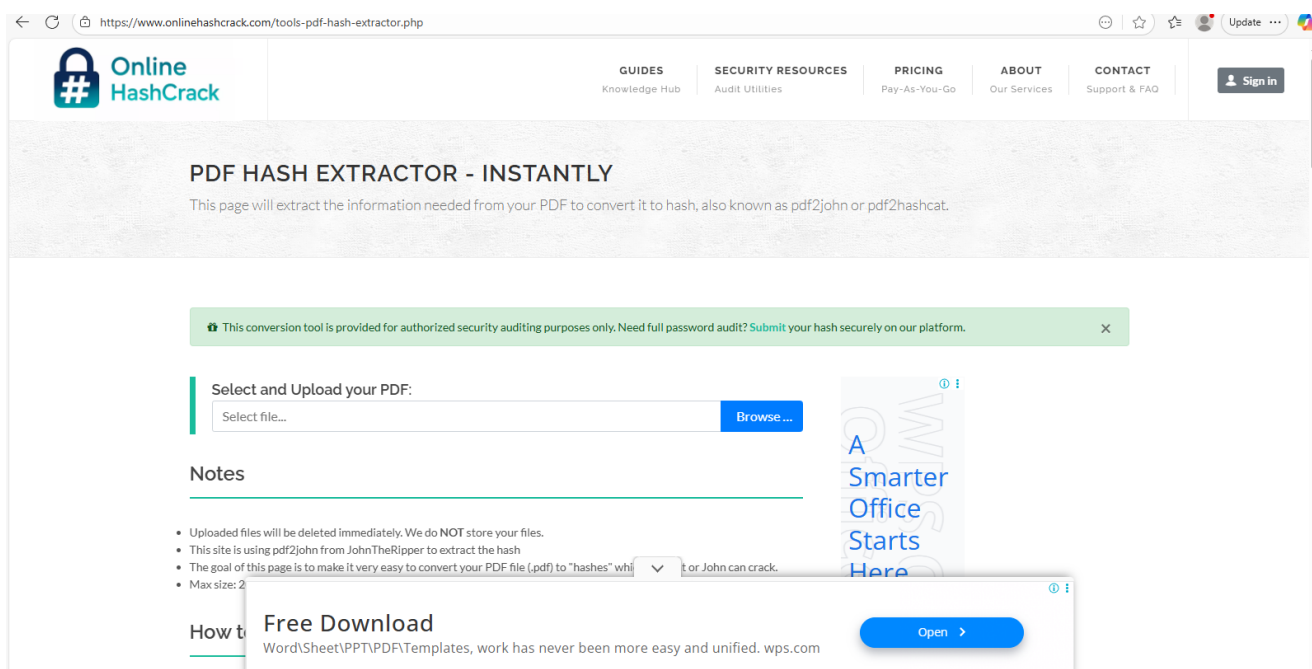
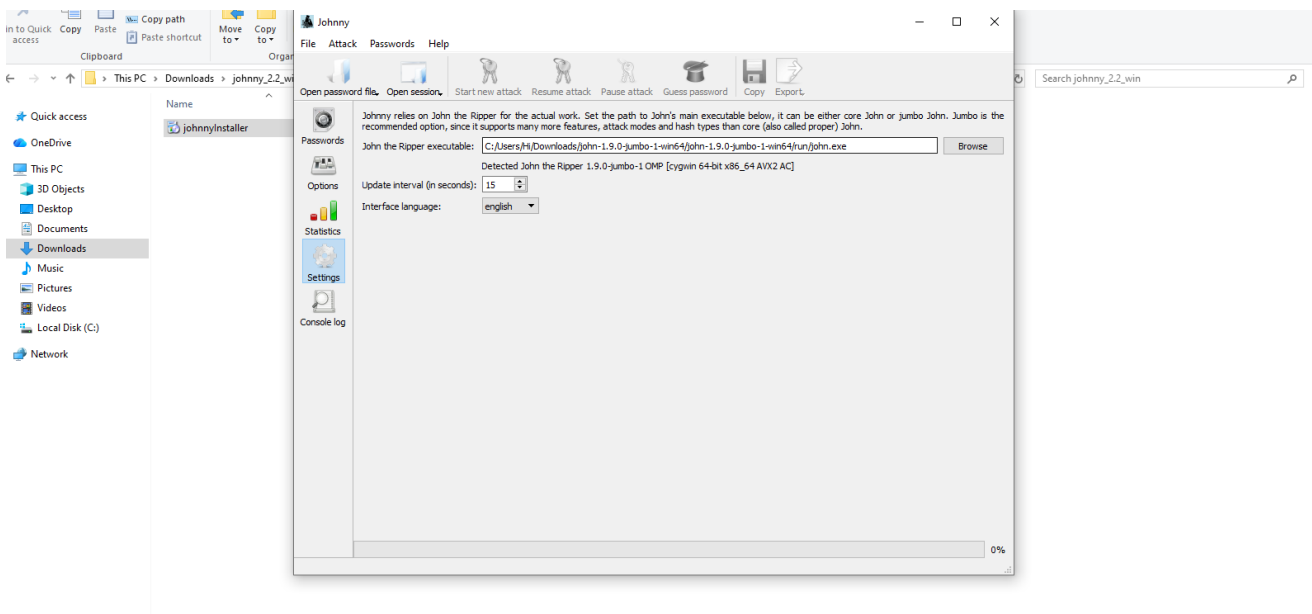
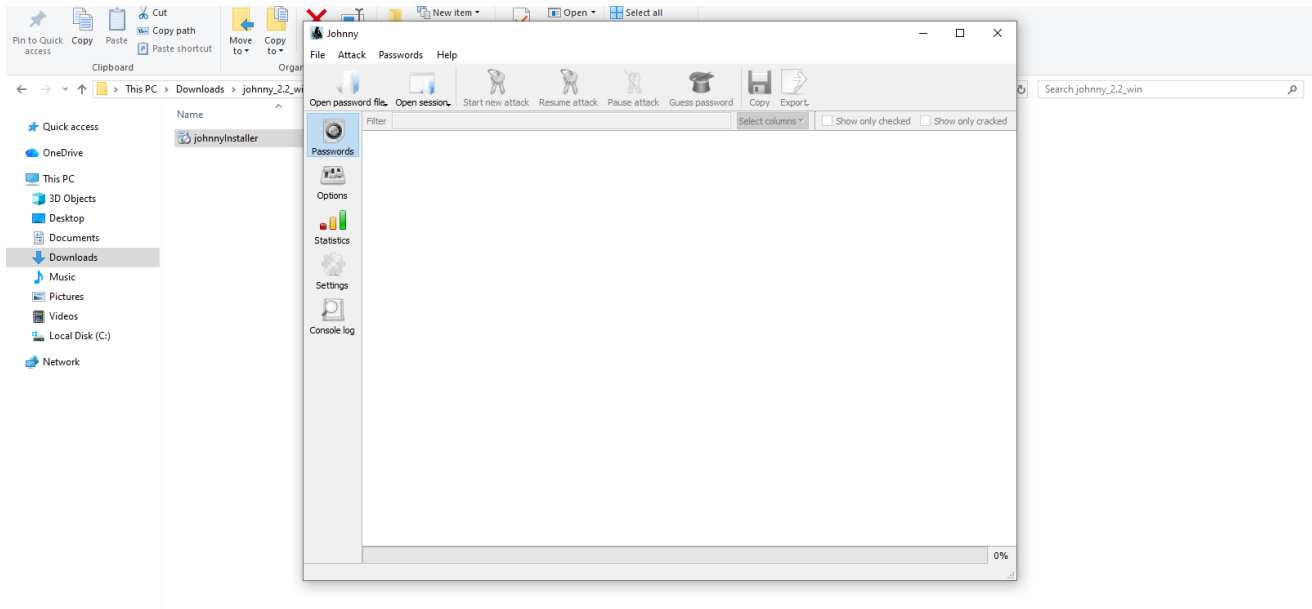
Deb packages (suitable for Debian, Linux Mint, Ubuntu and so on):

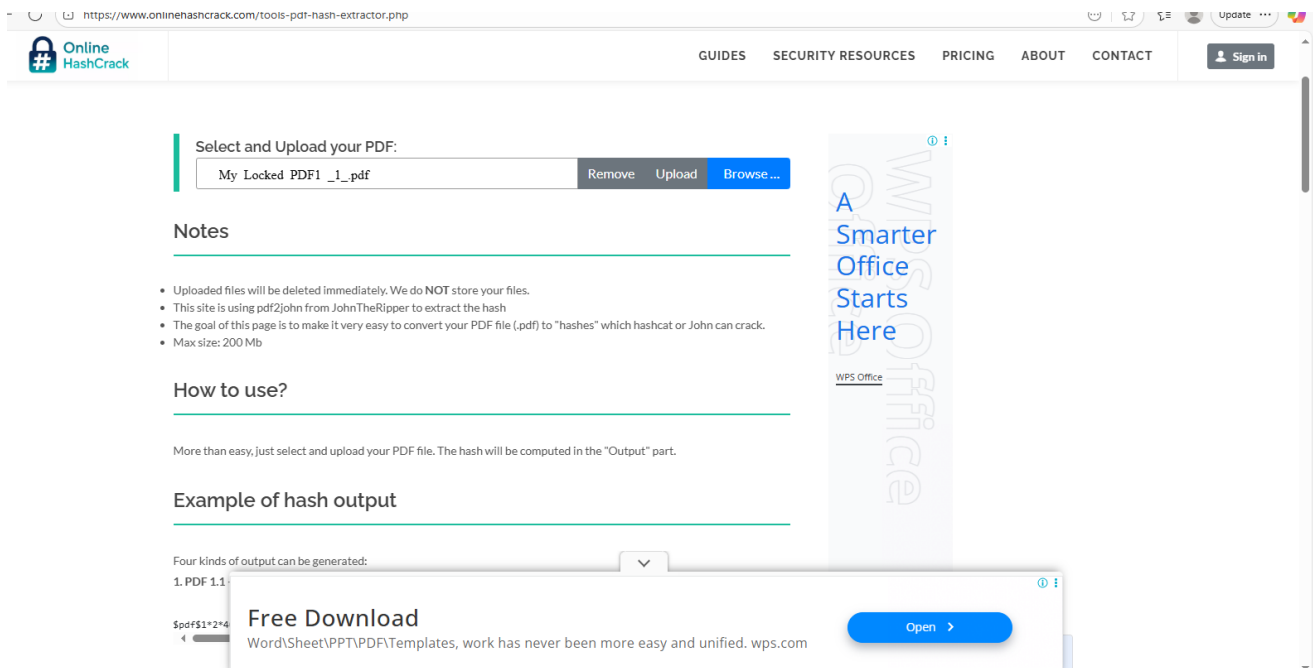
- x86 (i386, i586) architecture: [johnny_1.1.3_i386.deb](#),
- x86_64 (amd64) architecture: [johnny_1.1.3_amd64.deb](#).

Rpm packages (suitable for Fedora, Mageia, OpenSUSE):

- x86 (i386, i586) architecture: [johnny-1.1.3-1.i586.rpm](#),
- x86_64 (amd64) architecture: [johnny-1.1.3-1.x86_64.rpm](#).

Generic tarball ([manual installation](#) or no installation):





3. Results

Metric

Result

Completion

Passwords cracked

Remaining

100%

1/1

0

Hash format

4. Verification

PDF

The recovered password was successfully used to unlock the authorized training PDF. The password itself is intentionally omitted from the portfolio report.

5. Learning Outcomes

Practical experience was gained with password hashes, John the Ripper, Johnny, PDF hash extraction, controlled password-recovery workflows, and ethical security testing.

6. Ethical Considerations

Password-cracking tools should only be used on files and systems for which explicit authorization has been provided.